

## Interim Report on UESTC4006P(BEng) Final Year Project

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|                                                              |                                                                                                                                                                                                  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Placement Company<br>(if appropriate)                        | N/A                                                                                                                                                                                              |
| Working Title of Project                                     | The implementation of gesture recognition algorithm software based on wearable system                                                                                                            |
| Name of First Supervisor                                     | Ma Min                                                                                                                                                                                           |
| <b>Declaration of Originality and Submission Information</b> | <p><i>I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements</i></p> <p>Signed (Student) : 颜瑾溪</p> |

### 1. Project Outline

#### 1.1 Project Background and Motivation

Surface Electromyography (sEMG) has emerged as a pivotal technology in the field of Human-Machine Interaction (HMI), particularly for prosthetic control and wearable gesture recognition devices 错误!未找到引用源。错误!未找到引用源。 Unlike computer vision-based solutions, sEMG captures the electrical activity generated by skeletal muscles directly from the skin surface, offering a privacy-friendly, non-optical, and highly portable alternative for intention detection [3]. However, the practical deployment of sEMG-based systems faces significant engineering challenges. sEMG signals are stochastic, non-stationary, and highly susceptible to various environmental noises such as power line interference, motion artifacts, and electrode displacement[4]. While existing literature often focuses on maximizing accuracy in controlled laboratory environments, there is a lack of systematic evaluation regarding the robustness of these systems under varying noise conditions. This project is motivated by the need to bridge the gap between high theoretical accuracy and practical reliability in real-world scenarios.

#### 1.2 Project Objectives and Scope

The primary objective of this project is to develop a robust machine learning pipeline for sEMG-based gesture recognition and to systematically evaluate its performance under simulated noise conditions. Specifically, the project aims to construct a reproducible processing pipeline encompassing data preprocessing, feature extraction, and classification using the NinaPro DB5 dataset. A key performance target is to achieve a baseline classification accuracy of at least 85% on the clean dataset using computationally efficient algorithms. Furthermore, the study seeks to quantify the degradation of system performance when subjected to various types of noise, such as Gaussian noise and motion artifacts, at different intensity levels. The current scope focuses on intra-subject classification using classical machine learning models to establish a solid baseline, while deep learning approaches and inter-subject generalization are reserved for future phases.

#### 1.3 Overall Methodology Overview

The project follows a modular engineering pipeline designed to ensure data integrity and reproducibility. As illustrated in Figure 1-1, the workflow proceeds systematically from raw signal

acquisition through segmentation, feature extraction, and finally to classification and robustness testing.

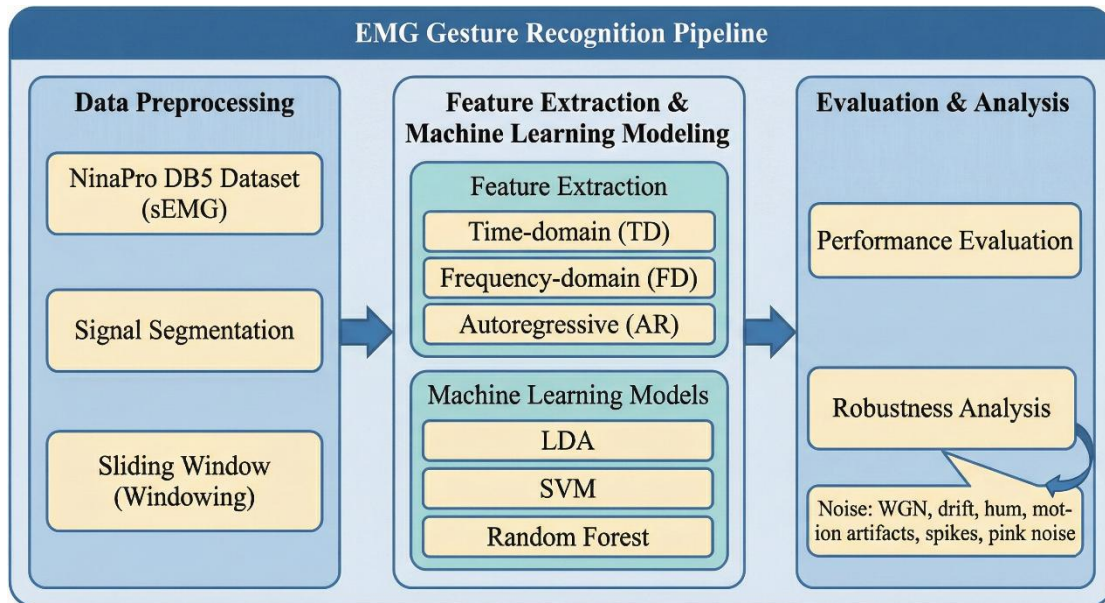


Figure 1-1 Overall experimental pipeline of the proposed EMG-based gesture recognition system.

This structured approach allows for the precise isolation of variables, enabling a clear analysis of how specific processing stages contribute to overall system performance. A distinct feature of this methodology is the inclusion of a "Noise Injection Module," which facilitates the controlled addition of synthetic noise to clean data for rigorous stress-testing of the models.

## 2. Work Done So Far (Methods, Models, Experiments)

### 2.1 Dataset and Signal Preprocessing

This study utilizes the NinaPro DB5 dataset, a recognized benchmark in the field of prosthetic control[1]. The dataset was selected due to its high resolution and double-myoelectric armband configuration, which provides a comprehensive mapping of forearm muscle activity. The data includes recordings from 10 intact subjects performing a wide range of hand gestures. For this interim report, the focus is placed on a subset of 17 basic gestures (Exercise B), sampled at 200 Hz with 16 active channels. The experimental methodology is governed by a strict processing pipeline to ensure consistency.

To translate continuous EMG streams into discrete classification inputs, a sliding window technique was employed. A window length of 200 ms was selected to capture the minimum time required for a distinct muscle activation pattern, while a step size of 50 ms (150 ms overlap) was used to augment the dataset and ensure temporal continuity. As depicted in Figure 2-1, this overlapping strategy significantly increases the

sample density and smooths the transition between the resting state and active muscle contraction.

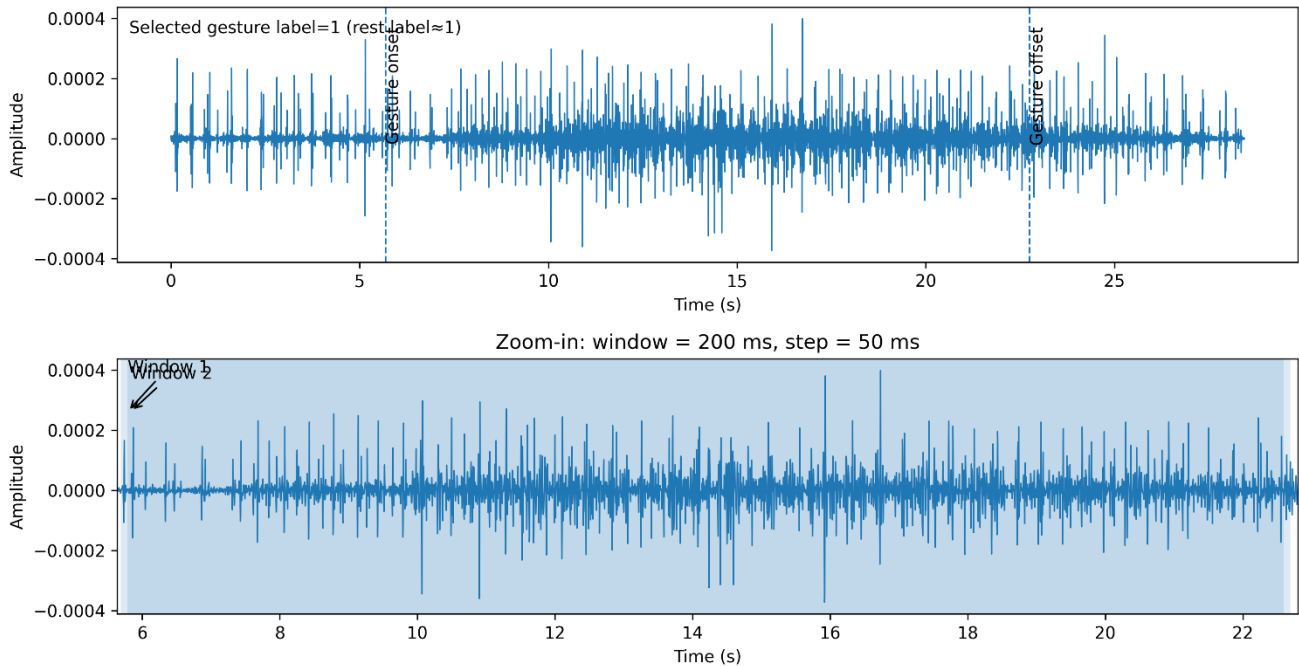


Figure 2-1 Example of EMG Segmentation and Windowing.

## 2.2 Feature Extraction Methodology

To address the stochastic nature of raw EMG signals, a hybrid feature set was extracted from each time window. The feature vector was constructed in three stages. First, Time-domain (TD) features were calculated, including Mean Absolute Value (MAV), Zero Crossing (ZC), Slope Sign Change (SSC), and Waveform Length (WL). These features capture the energy and frequency information without requiring complex transformation, as originally proposed by Hudgins et al. [5]. Second, Frequency-domain (FD) features, such as Mean Frequency (MNF) and Median Frequency (MDF), were derived via Fast Fourier Transform (FFT) to capture the spectral characteristics of the signal [6]. Finally, 4th-order Autoregressive (AR) coefficients were calculated to model the signal as a linear combination of its previous values. A feature ablation study, detailed in Section 2.5, was conducted to validate the necessity of this high-dimensional feature set.

## 2.3 Machine Learning Models

Three distinct classifiers were implemented to establish a performance baseline: Linear Discriminant Analysis (LDA), Support Vector Machine (SVM), and Random Forest (RF). LDA was selected for its simplicity and low computational cost, while SVM was implemented with a Radial Basis Function (RBF) kernel to handle non-linear decision boundaries. The Random Forest model[7] was included for its inherent

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resistance to overfitting and capability to handle high-dimensional data[7]. All models were evaluated using a 5-fold cross-validation protocol to ensure the statistical reliability of the results.

## 2.4 Baseline Performance Results

The initial experiments focused on establishing a "clean baseline" using the unprocessed NinaPro DB5 data. As shown in Table 2-1, the Random Forest (RF) model outperformed both LDA and SVM, achieving a mean accuracy of  $92.08\% \pm 0.47\%$  and a Macro F1-score of 0.9207. This performance significantly exceeds the project target of 85%. The low standard deviation indicates that the RF model is stable across different folds of the data. Consequently, Random Forest was selected as the primary model for the subsequent robustness analysis.

| Model         | Accuracy (mean $\pm$ std) | Macro F1 (mean $\pm$ std) |
|---------------|---------------------------|---------------------------|
| LDA           | $0.3770 \pm 0.0081$       | $0.3760 \pm 0.0079$       |
| SVM (RBF)     | $0.7259 \pm 0.0041$       | $0.7267 \pm 0.0039$       |
| Random Forest | $0.9208 \pm 0.0047$       | $0.9207 \pm 0.0047$       |

Table 2-1 Baseline Model Performance Comparison.

## 2.5 Robustness Analysis and Experimental Results

To evaluate system reliability, various noise types were synthesized and injected into the test sets. Figure 2-2 and Figure 2-3 visualize the characteristics of these injected noises. While White Gaussian Noise (WGN) distributes energy across the spectrum, Baseline Drift primarily affects low frequencies, potentially confusing features like Mean Absolute Value.



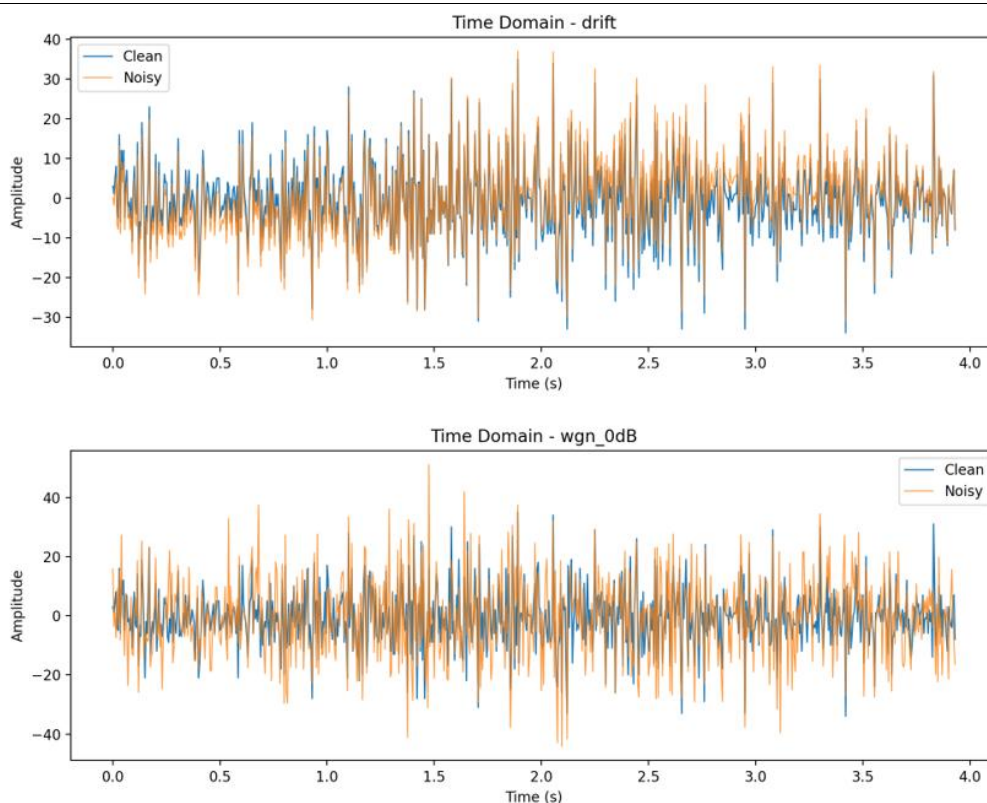


Figure 2-2 Noise Injection Example (Time Domain).

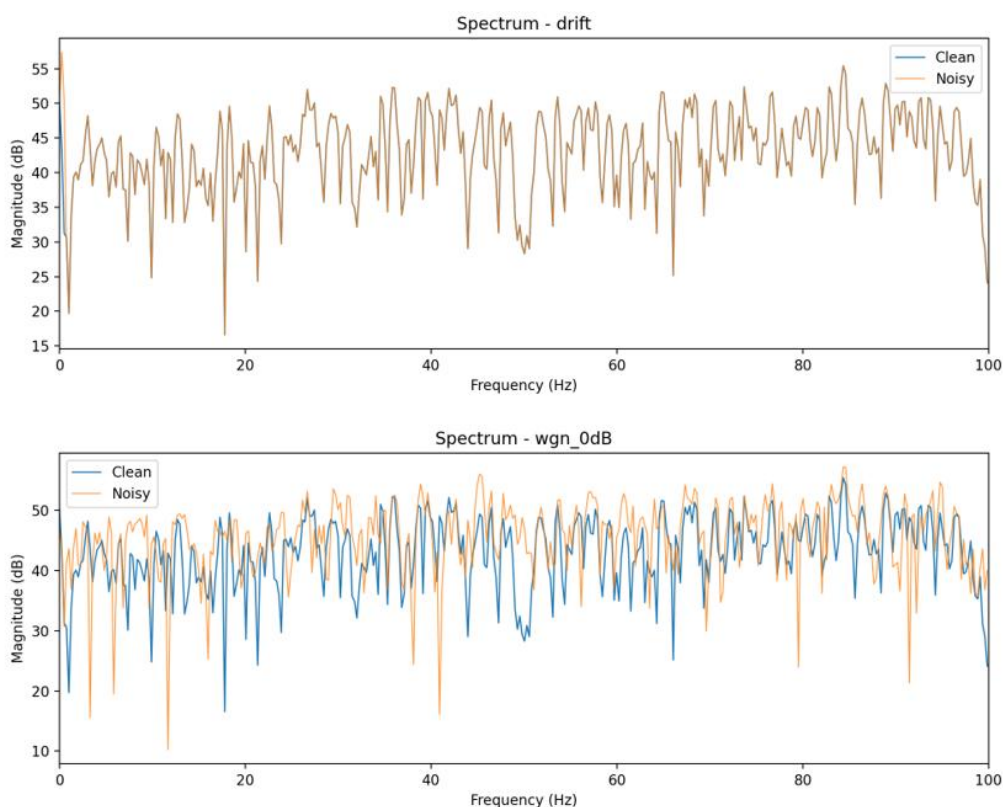


Figure 2-3 Noise Injection Example (Frequency Domain).

To verify the efficiency of the feature set, an ablation study was conducted. Figure 2-4 demonstrates a diminishing return in performance. The transition from TD to TD+FD yielded a significant accuracy boost. However, adding AR features (TD+FD+AR) provided a smaller marginal gain, suggesting that the feature space is becoming saturated and potentially redundant.

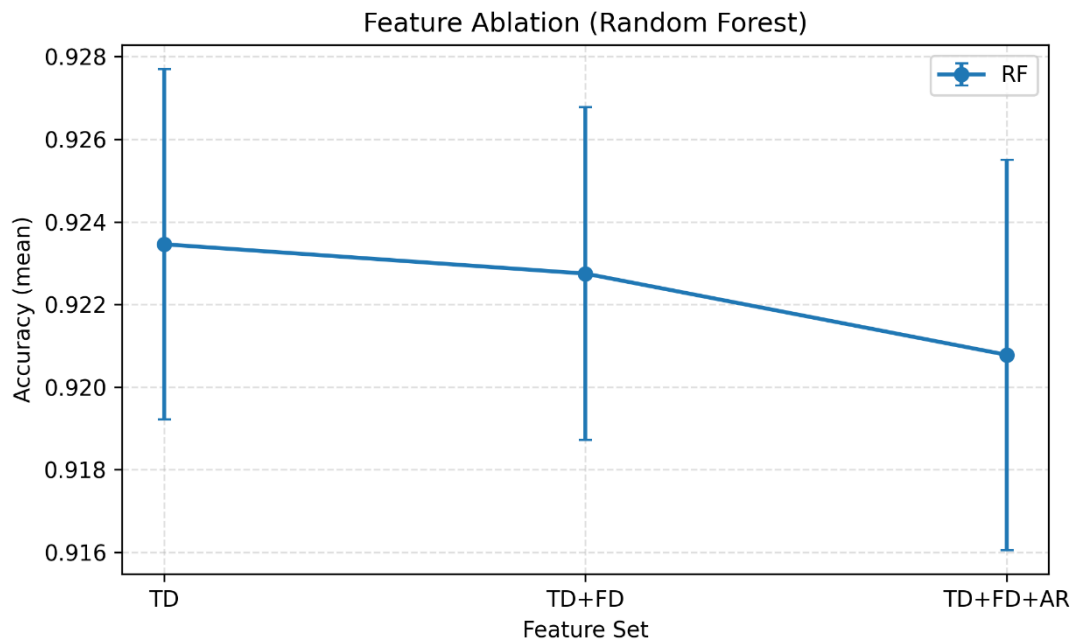


Figure 2-4 Feature Ablation Study showing accuracy gains with feature set expansion.

The core of the experimental work involved testing the RF model against increasing noise levels. Figure 2-5 illustrates the degradation curves. The system maintains robustness at high SNR (>20dB) but degrades rapidly under intense Motion Artifacts. Figure 2-6 provides a summary view, highlighting that while the clean accuracy is high (~92%), the average performance drops significantly when subjected to realistic noise types, particularly motion-based disturbances.

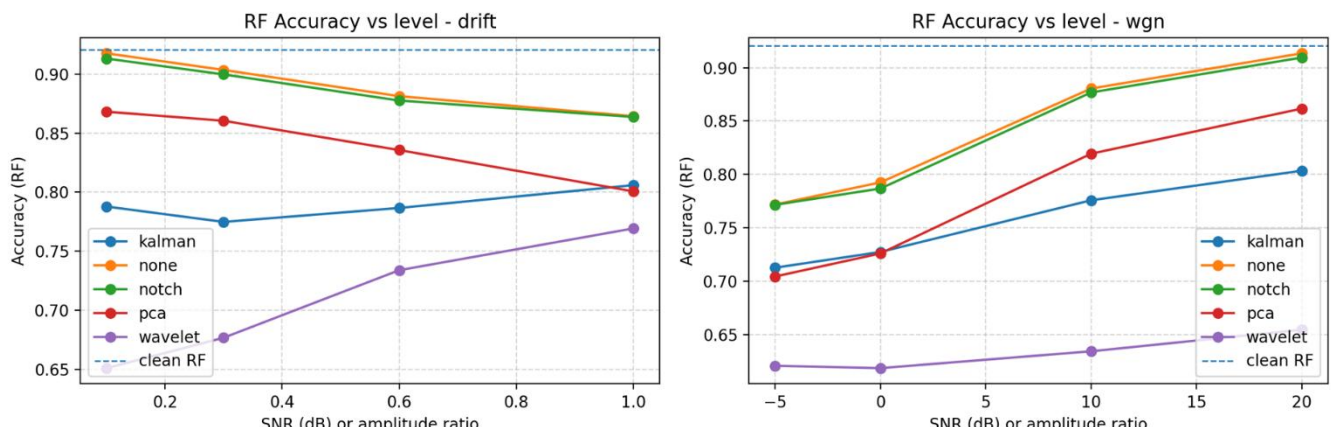


Figure 2 错误!文档中没有指定样式的文字。 -5 Accuracy vs Noise Level (Signal-to-Noise Ratio) for WGN and Motion Artifacts.

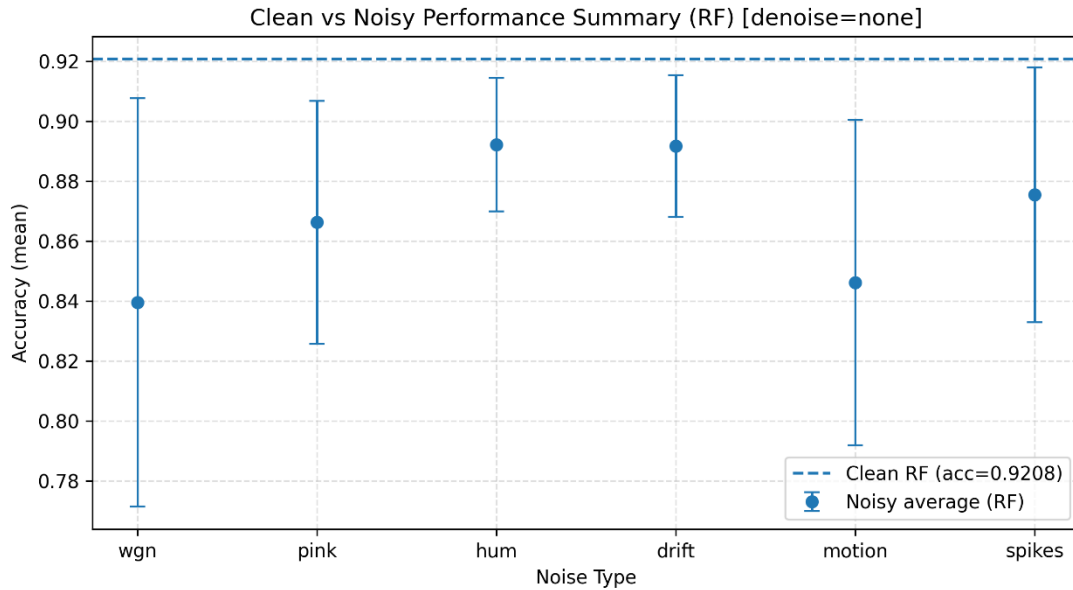


Figure 2-6 Clean vs Noisy Performance Summary (RF).

### 3. Challenges and Solutions

#### 3.1 Challenge: High-Dimensional Feature Redundancy

The experimental results presented in Chapter 2 indicated that while stacking Time-domain, Frequency-domain, and Autoregressive features maximizes accuracy, it leads to a high-dimensional feature vector of over 100 dimensions per window. This "curse of dimensionality" introduces computational latency and increases the risk of overfitting, particularly when the model is exposed to data outside the training distribution. The marginal gain observed from adding AR coefficients suggests that significant information redundancy exists within the current feature set. To address this, future work will implement Dimensionality Reduction techniques. Specifically, methods such as Principal Component Analysis (PCA) or feature selection strategies discussed by Phinyomark et al. will be applied to project the features into a lower-dimensional orthogonal space[6]. This solution aims to maintain the accuracy above 85% while reducing the computational load by at least 30%.

#### 3.2 Challenge: Sensitivity to Non-Stationary Noise

The robustness analysis revealed a critical vulnerability where the model's performance degrades non-linearly under specific noise conditions, particularly Motion Artifacts and Baseline Drift. Unlike Gaussian noise, which is uniform, motion artifacts mimic the frequency content of actual muscle contractions, creating low-frequency transients. The current preprocessing pipeline, which relies on standard filtering,



fails to distinguish between voluntary muscle movement and artifact-induced spikes. To mitigate these effects, the proposed solution involves implementing Adaptive Filtering and Data Augmentation. Specifically, replacing fixed filters with adaptive algorithms, such as those discussed by Sensinger et al., allows for the tracking and cancellation of systematic noise like power line interference[8]. Furthermore, the training dataset will be augmented with synthetic noise profiles generated in Chapter 2. By exposing the Random Forest model to "noisy" examples during the training phase, the model is expected to learn to ignore artifact-specific features, thereby improving its generalization capability in real-world scenarios.

### 3.3 Security and Quality Assurance

From a system engineering perspective, ensuring the integrity of the data and the reproducibility of the results is paramount. In a potential real-world deployment, sEMG data could be subject to adversarial attacks, such as injected signal patterns designed to force specific gestures. To align with ISO 9001:2015 quality management principles, the project adopts a strict Process Approach involving comprehensive version control and logging. All data processing scripts and model parameters are version-controlled, and every experiment generates a log file recording the exact hyperparameters and random seeds. Although this is a research project, the "Noise Injection" methodology serves as a proxy for adversarial robustness testing. By evaluating how the model behaves under extreme noise, the study effectively assesses its resilience against signal jamming or data corruption, ensuring the system fails gracefully rather than producing hazardous false positives.

## 4. Conclusions from Initial Work and Future Work

### 4.1 Conclusions from Initial Experimental Results

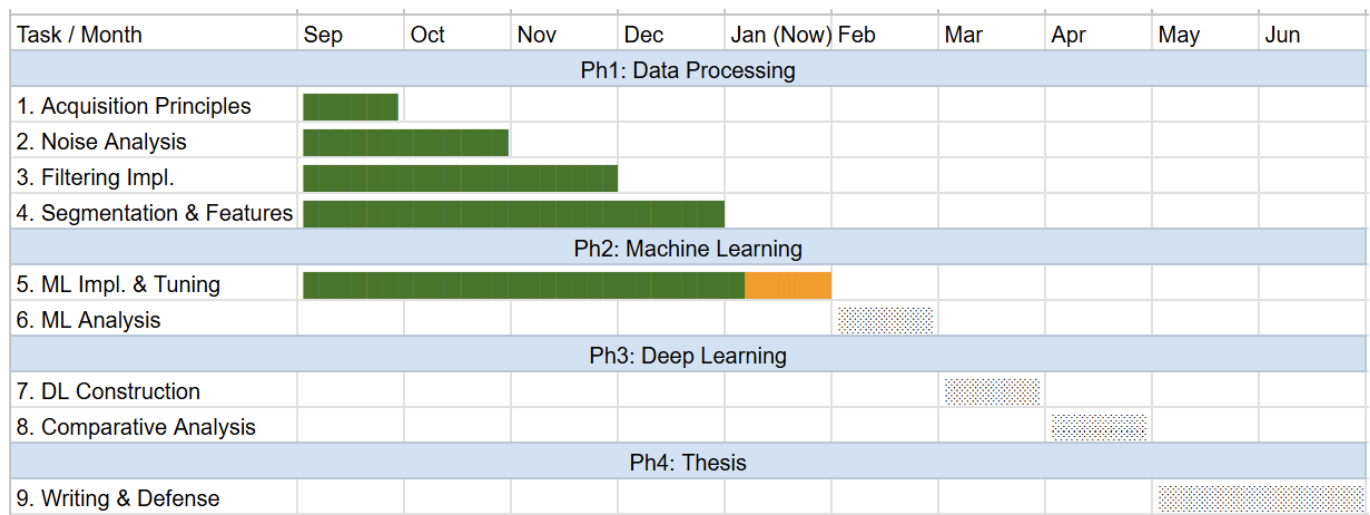
The work conducted in the first phase of this project has successfully established a high-performance, reproducible pipeline for sEMG gesture recognition. First, the Random Forest classifier has proven to be a highly effective baseline, achieving an average accuracy of 92.08% on the clean dataset. This result not only meets but significantly exceeds the initial project milestone of 85%, confirming the validity of the chosen segmentation and feature extraction strategies. Second, the combination of Time-domain and Frequency-domain features was found to provide the most cost-effective performance, while Autoregressive features introduced redundancy that must be managed in future iterations. Finally, the systematic noise injection study revealed that the system is relatively robust to white noise but highly sensitive to structural disturbances like motion artifacts and baseline drift. This finding defines the critical path for the next phase of development: robustness is not about handling static, clean data, but about

surviving dynamic, noisy environments.

### 4.2 Planned Improvements and Future Work

Building upon the established baseline and the identified challenges, the second phase of the project will focus on several key areas. Future improvements will include advanced feature engineering, specifically the implementation of PCA or LDA for dimensionality reduction to optimize the trade-off between accuracy and computational efficiency. Additionally, the project will explore Deep Learning approaches, such as Convolutional Neural Networks (CNNs), to accept raw sEMG spectrograms as input[9]. This aims to bypass manual feature extraction and potentially learn more robust, invariant representations of the muscle signals. The "Training with Noise" approach proposed in Section 3.2 will also be executed, where the model will be retrained on a mixed dataset containing both clean and noisy samples to empirically verify if robustness can be learned. Finally, initial tests will be conducted to evaluate the model's performance on inter-subject tasks, addressing the challenge of user variability.

### 5. Revised Gantt Chart



### 6. References

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